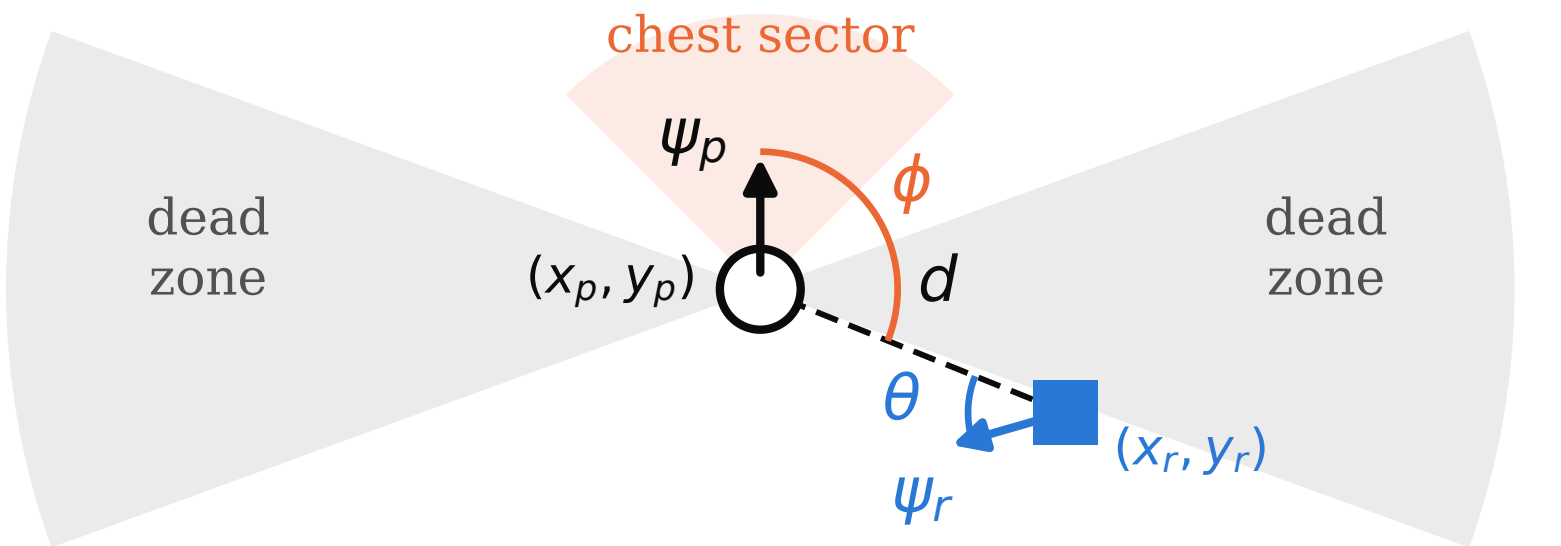
chest sector
ψ_p
ϕ
dead
zone
(x_p, y_p)
d
dead
zone
θ
(x_r, y_r)
ψ_r